Grafana 13.3 Cheat Sheet

Dashboards, alerting and as code. Lab: Docker Compose grafana profile (Alloy, Prometheus, Loki, Tempo).

DATA FLOW

- Dashboard: data source → query → transformation → panel (field config) → dashboard (variables, annotations, links).
- Alert rule: queries → expressions (Reduce, Math, Threshold) → condition → evaluation group → alert instance → notification policy → contact point.
- Explore for ad-hoc queries; Drilldown apps for queryless exploration.
- Collect with Alloy (Agent and Promtail are end of life); store in Prometheus or Mimir, Loki, Tempo.

DOCKER RUN AND GF_ ENV VARS

```
docker run -d -p 3000:3000 \
 --name=grafana \
 -v grafana-storage:/var/lib/grafana \
 -e GF_SERVER_ROOT_URL=\
   http://localhost:3000/ \
 -e GF_PLUGINS_PREINSTALL=\
   grafana-clock-panel \
 grafana/grafana-enterprise:13.3.0
```

- Pattern `GF_<SECTION>_<KEY>`, upper case; env beats `custom.ini` beats `defaults.ini`.
- `GF_SECURITY_ADMIN_PASSWORD__FILE` reads a secret from a file.
- `GF_METRICS_ENABLED`, `GF_LOG_LEVEL`, `GF_FEATURE_TOGGLES_<name>`.
- `GF_PATHS_PROVISIONING=/etc/grafana/provisioning`, data `/var/lib/grafana`.

PROVISIONING FOLDERS

```
provisioning/
  datasources/*.yaml
  dashboards/*.yaml (providers)
  alerting/*.yaml
```

- Mount read-only at `/etc/grafana/provisioning/...`; provisioned items are read-only in the UI.
- Reload without restart: `POST /api/admin/provisioning/alerting/reload` and `.../datasources/reload`.

VARIABLES AND BUILT-INS

Types: Query, Custom, Text box, Constant, Data source, Interval, Filters, Switch.
Syntax: `$var`, `${var}`, `${var:format}`.

regex	`(a|b)` for =~ matchers
pipe	`a|b`
csv	`a,b`
lucene	`("a" OR "b")`
json	`["a","b"]`
queryparam	`var-x=a&var-x=b`
text / raw	label / no escaping

Built-ins: `$__interval`, `$__rate_interval`, `$__range`, `$__range_s`, `$__from`, `$__to`, `$__dashboard`, `${__user.login}`.

- Use `$__rate_interval` for `rate()` on Prometheus.

VISUALIZATION PICKER

Trend	Time series
Now	Stat; Gauge, Bar gauge
Distribution	Histogram, Heatmap
Parts of whole	Pie, Bar chart
State over time	State timeline, Status history
Top N, exact	Table, sorted Bar gauge
Where	Geomap
Related	XY chart, Trend, Node graph
Logs, traces	Logs, Traces, Flame graph
Context	Text, Dashboard list, Alert list

FIELD CONFIG ORDER

Standard options (unit, min/max, decimals, display name, colour) → Thresholds → Value mappings → Overrides.

- Defaults apply to all fields; overrides (matcher + properties) win on selected fields.
- Clicking a legend entry to isolate a series creates an override.
- Thresholds colour values and drive Stat, Gauge and Table cells.

ALERT RULE ANATOMY

Queries (A, B) → Expressions (Reduce, Math, Threshold) → **Condition** → Evaluation group (interval) → **Pending period** → Alerting → **Keep firing for**.

- Pending period 0 skips Pending; keep firing for 0 skips Recovering.
- Pending is counted in evaluations: first notification arrives between 5 and 6 min for a 1 min interval and 5 min pending.
- Set No Data and Error handling per rule; labels route, annotations explain (`summary`, `runbook_url`).

ALERT INSTANCE STATES

```
Normal -> Pending -> Alerting
Alerting -> Recovering -> Normal
Pending -> Normal (clears early)
No Data / Error (own states,
  or mapped to Normal/Alerting)
```

- Resolved message is sent when an instance returns to Normal; it waits for the group interval.
- State history is stored in Loki, label `from="state-history"`.

NOTIFICATION POLICY

- Matchers: `=`, `!=`, `=~`, `!~` on labels; first match wins unless Continue matching.

group_wait	`30s`
group_interval	`5m`
repeat_interval	`4h`

- Group by `alertname` (and `grafana_folder`); `DatasourceNoData` and `DatasourceError` alerts can get their own route.
- Attach mute timings or active time intervals to child policies.

CONTACT POINT TYPES

People/chat	Email, Slack, Teams, Telegram, Discord, Google Chat, Webex, Line, DingDing, WeCom, Threema, Pushover
On-call	PagerDuty, Opsgenie, VictorOps, Sensu Go, Grafana IRM
Tickets	Jira, Amazon SNS, Kafka REST Proxy, MQTT
Generic	Webhook, Alertmanager

- Email needs SMTP (`enabled`, `host`) set on the server.

GO TEMPLATE SNIPPET

```
{{ define "bk.title" -}}
[{{ .Status | toUpper }}] {{
.GroupLabels.alertname }}
{{- end }}
{{ define "bk.msg" -}}
{{ range .Alerts.Firing }}
- {{ .Labels.alertname }}: {{
.Annotations.summary }}
  {{ .DashboardURL }} {{ .SilenceURL }}
{{ end }}{{- end }}
```

- Use `{{ template "bk.msg" . }}` in the integration; names are unique across groups.
- Empty `.GroupLabels.alertname` or `runbook_url` fail silently.

SILENCE VS MUTE VS INHIBITION

Silence	One-off, start and end, label matchers
Mute timing	Recurring schedule on a child policy
Active interval	Notify only inside the schedule
Inhibition	Source firing suppresses target; `inhibitionrules` in `/apis/notifications.alerting.grafana.app/v1beta1`

- Rules keep evaluating; only notifications stop.

RECORDING RULES

- Pre-compute an expensive query and write it back to a Prometheus-compatible data source.
- Set `metric`, `from` and `target_datasource_uid`; no condition, `noDataState` or `execErrState`.
- Name `level:metric:operations`.

AS-CODE OPTIONS

File	`provisioning/` YAML
Git Sync	bidirectional, pull requests
Foundation SDK	typed dashboards (Go, TS, Python, PHP, Java)
Terraform	`grafana_rule_group`, `grafana_contact_point`, `grafana_notification_policy`, `grafana_message_template`, `grafana_mute_timing`
Operator	Kubernetes custom resources
gcx	CLI, replaces grafanactl

REMOVED / RENAMED

Legacy alerting	removed in 11.0
Angular panels	removed in 12; Graph is Time series
API keys	service accounts
Grafana Agent, Promtail	Alloy
Public dashboards	externally shared dashboards
Grizzly, grafanactl	`gcx`